

General Features

Sensor	Internal 5Hz high sensitivity geophone Internal 10Hz high sensitivity geophone Optional KCK version for external string
Digitalization	24-bit $\Delta\Sigma$ ADC with GNSS disciplined clock
Storage Capacity	16GB (32GB optional)
Node Autonomy	Internal 52/60WH lithium-ion battery pack supports 28/33 days continuous recording
Timing Accuracy	$\pm 10\mu\text{s}$ (GNSS disciplined) $\pm 1\text{ms}$ (within 1 hour without GNSS signal)
Automated Tests	Internal testing signal generator for function and performance self-inspection and diagnosis
Working Mode	Autonomy with wireless on-site QC inspection, Support UAV patrol
LED Status Indicator	Nodal health, GNSS synchronization status, Wireless data harvesting status
Data Harvest	Charging/Data download rack
Infield Management	Handheld android tablet
Weight	750g (excl. spike), 800g (incl. spike)
Size	9.8cm x 10.8cm x 11cm (excl. spike)
Operating Temperature	-40°C ~ +70°C

Acquisition Specifications

ADC Resolution	24-bit
Sample Interval	0.5ms, 1ms, 2ms, 4ms
Preamplifier Gain	x1, x4, x16 (0dB, 12dB, 24dB)
Input Impedance	40K Ω 22nF (5Hz geophone) 20K Ω 22nF (10Hz geophone)
Gain Accuracy	0.1%
Full Scale Input Signal	$\pm 2.5\text{Vp-p}$ @ x1 gain $\pm 625\text{mVp-p}$ @ x4 gain $\pm 156\text{mVp-p}$ @ x16 gain
Instantaneous DR @1ms(typical value)	125dB @ x1 gain
Full Scale DR	145dB
Equivalent Input Noise @1ms(typical value)	1 μV @ x1 gain, 0.25 μV @ x4 gain
Harmonic Distortion	-120dB @ 31.5Hz
Frequency Response	0Hz ~ 866Hz @ 0.5ms (-3dB) 0Hz ~ 800Hz @ 0.5ms (-0.1dB)
Anti-alias Filter	Linear filter, 86.6% of Nyquist -3dB bandwidth
Stopband Attenuation	>120dB @Nyquist
Common Mode Rejection	>120dB

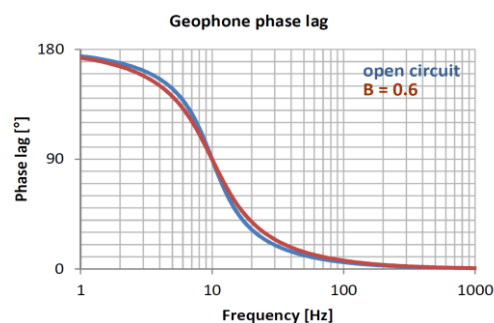
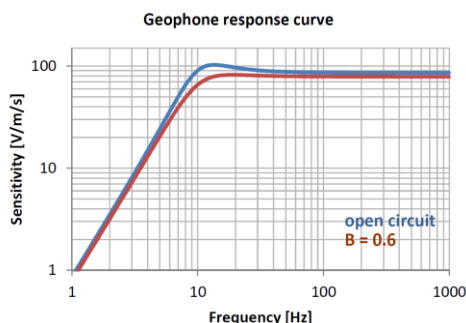
Product Highlights

- Built-in 5Hz (5Hz~150Hz) or 10Hz (10Hz~240Hz) high sensitivity geophone;
- Optional KCK version for external geophone string;
- 24-bit $\Delta\Sigma$ ADC converter for high resolution, high dynamic range seismic signal recording;
- Configurable sample rate and preamplifier gain;
- High sensitive GNSS module equipped with active patch antenna supports GPS, Beidou, and GLONASS;
- Intermittent GNSS clock synchronization strategy, with timing accuracy less than $\pm 10\mu\text{s}$;
- Internal lithium-ion battery pack supports seismic recording for up to 800 hours;
- Built-in test signal generator supports on-site full signal chain function and performance self-inspection and diagnosis, including system noise, real-time dynamic range, geophone impedance, natural frequency, sensitivity, damping, distortion, etc.;
- BLE module together with visual LEDs for onsite configuration and proximity. Ground communication distance around 20 meters, air communication distance over 100 meters, supports vehicle or UAV patrol;
- P&R (Place & Record) mode for rapid and convenient infield deployment, supporting automatic GNSS position matching with line and point number;
- Industrial Android tablet supports working status verification after deployment, including battery autonomy, environmental temperature, tilting, ambient noise, real-time seismic waveform, etc.;
- High strength engineering plastic enclosure for enhanced anti-dropping capability, waterproof (IP68), flame retardant (5VA), ultraviolet aging (F1);
- Compact volume (9.8L x 10.8W x 11H cm) and light weight (750g) for convenient transportation and deployment.



10Hz High Sensitivity Geophone

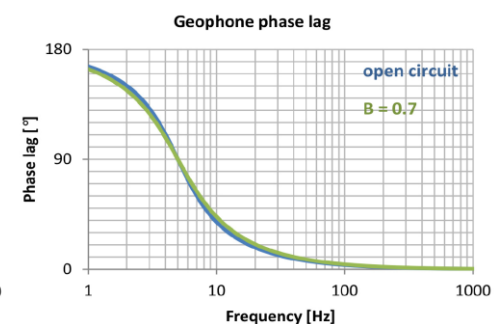
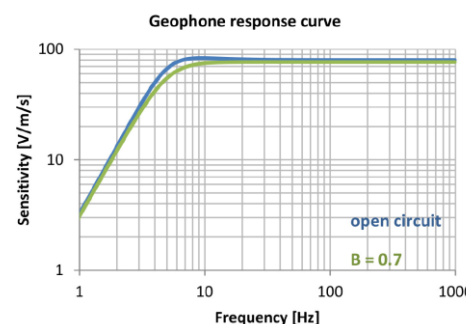
Natural Frequency	10Hz
Sensitivity	85.8V/m/s
Spurious	>240Hz
Distortion	≤0.1%
Coil Resistance	1800Ω
Tilt	±20°



10Hz High Sensitivity Geophone Response Curve and Phase Lag

5Hz High Sensitivity Geophone

Natural Frequency	5Hz
Sensitivity	80V/m/s
Spurious	>150Hz
Distortion	≤0.1%
Coil Resistance	1850Ω
Tilt	±10°



5Hz High Sensitivity Geophone Response Curve and Phase Lag

Charging Rack

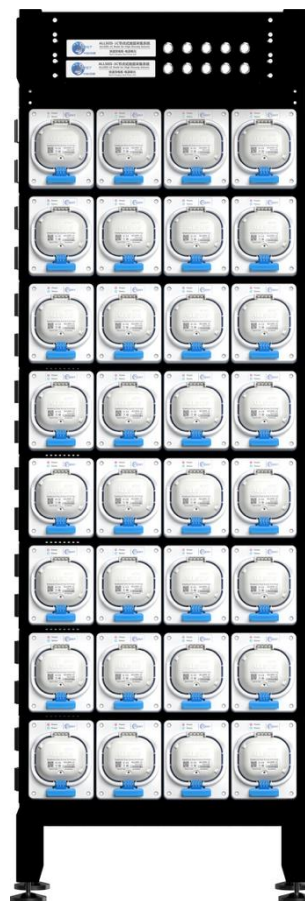
Slots	32
Charging Time	3.5 hours
Charging Power	700W
Unit Charging Power	18W
Operating Temperature	0°C ~ +40°C
Size	194x64x38cm

Data Download Rack

Slots	32
Unit Download Rate	0.6GB/min.
Full Download Rate	18GB/min.
Interface	10G SFP+
Operating Temperature	0°C ~ +40°C
Size	194x64x38cm

Desktop Data Download Rack

Slots	14
Unit Download Rate	0.6GB/min.
Full Download Rate	8GB/min.
Interface	2.5G Ethernet
Operating Temperature	0°C ~ +40°C
Size	94x64x38cm



ALLSEIS-1C:
Standard & KCK Version

Peripheral Equipment :
Data Management Server, Charging/Data Download Rack